



Lecture 05 Storage

Information Technology Fundamentals
IT-06016402

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Storage

Non and Volatile
หน่วยจำถาวรและชั่วคราว

volatile -> ไฟฟ้า
non-volatile -> ไม่จำเป็น

Non-Volatile
หน่วยจำถาวร เก็บข้อมูลถาวร สามารถถอดออกได้

Primary Storage	Secondary Storage	Offline Storage	Tertiary Storage
- It is the main memory where data and information are stored - It is called "memory" - Volatile/Non Volatile in nature - Examples are RAM, ROM, etc.	- It is referred to the external memory where data is stored <u>permanently</u> - Long-time storage - It is called "storage medium" - Non-volatile in nature - Examples are HDD, SSD, etc.	- It is called disconnected or removable storage. - It is a computer data storage on a medium or a device that is not under the control of a processing unit - It must be inserted or connected by a human operator before a computer can access it again. - Examples are CD DVD USB Flash Drive	- Similar to secondary storage, but can only be accessed via a network device. - Access is even slower. - It uses additional robotic mechanisms to transfer media between their long-term storage locations and available drives without human intervention. - Examples are NAS SAN – Tape Optical Disc

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Storage

สื่อ

A storage medium is the physical material on which a computer keeps data, information, programs, and applications

Server เก็บข้อมูลบนอินเทอร์เน็ต

Cloud storage keeps information on servers on the Internet, and the actual media on which the files are stored are transparent to the user

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Storage Device

- A storage device is the hardware that records and/or retrieves items to and from storage media



นำข้อมูลจากสื่อมาเก็บในหน่วยด.จ

Read

- Reading is the process of transferring items from a storage medium into memory

Input



Write

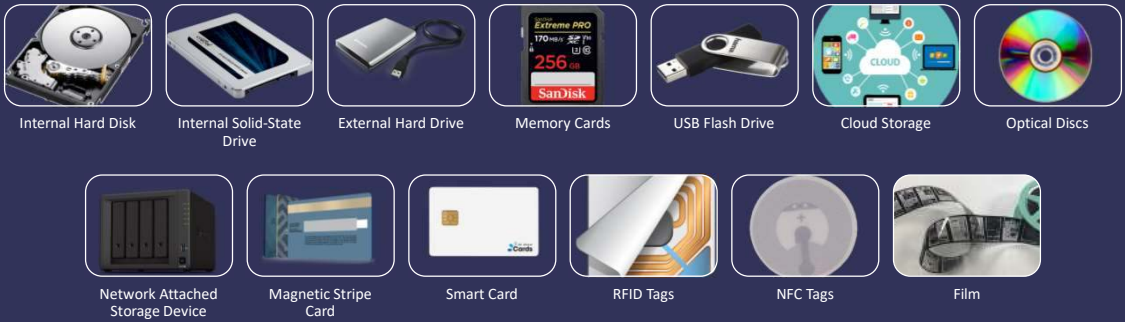
- Writing is the process of transferring items from memory to a storage medium

Output

It is not I/O device

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Storage



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Capacity

ความจุ

วัดประสิทธิภาพหน่วยจัดเก็บ
[ความจุ + เวลา]

- Capacity is the number of bytes a storage medium can hold

Storage Term	Short Form	Approximate Number of Byte	Exact Number of Byte
Kilobyte	KB	1 thousand (10^3)	$2^{10} = 1,024$
Megabyte	MB	1 million (10^6)	$2^{20} = 1,048,576$
Gigabyte	GB	1 billion (10^9)	$2^{30} = 1,073,741,824$
Terabyte	TB	1 trillion (10^{12})	2^{40}
Petabyte	PB	1 quadrillion (10^{15})	2^{50}
Exabyte	EB	1 quintillion (10^{18})	2^{60}
Zettabyte	ZB	1 sextillion (10^{21})	2^{70}
Yottabyte	YB	1 septillion (10^{24})	2^{80}

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Access Time

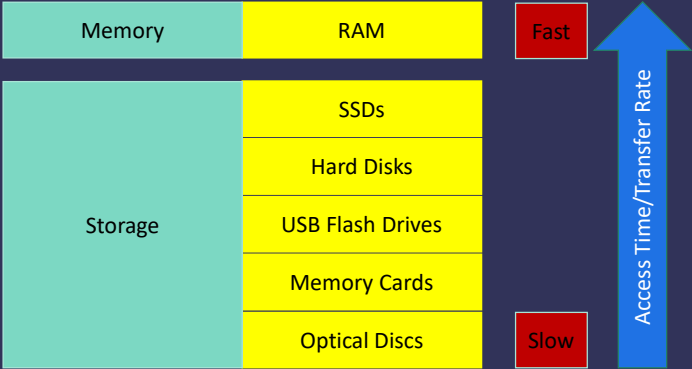
1 / transfer rate

ระยะเวลาที่ส่งข้อมูลจาก Memory ไปยัง Processor

วัดเป็น nanosecond.

- Access time measures:
 - The amount of time it takes a storage device to locate an item on a storage medium
 - The time required to deliver an item from memory to the processor
- Some manufacturers use **transfer rate**
 - Kbps (kilobyte per second)
 - Mbps (Megabyte per second)
 - Gbps (Gigabyte for second)

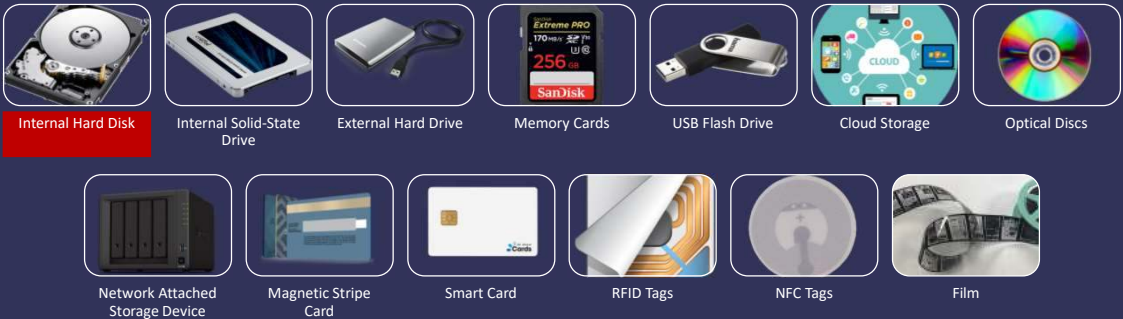
ค. 650 นี
ดูใกล้



Storage ที่อยู่ใกล้ CPU ก็จะเร็วจะดีกว่า

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Storage



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Hard Disk

ข้อมูลเล็กที่สุดที่คอมพิวเตอร์เก็บไว้
อนุภาคแม่เหล็กที่หน้ามาเรียงกันกับเป็น bit (ข้อมูล 0, 1)

- A hard disk, also called a hard disk drive (HDD) contains one or more inflexible, circular platters that use magnetic particles to store data, instructions, and information



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Hard Disk

ขนาดต่างกัน ขึ้นอยู่กับ 1) Platter (จำนวนจาน) เยอะ = จัดเก็บได้เยอะ
2) รูปแบบการจัดเก็บข้อมูลว่าใช้พื้นที่ที่เล็กที่สุดบนจาน
3) ความหนาแน่น

- The storage capacity of hard disks varies and is determined by:
 - The number of platters the hard disk contains
 - Whether the disk uses longitudinal or perpendicular recording
 - Density



ใช้วางตามแนวนอน
Longitudinal



ใช้วางตามแนวตั้ง
Perpendicular

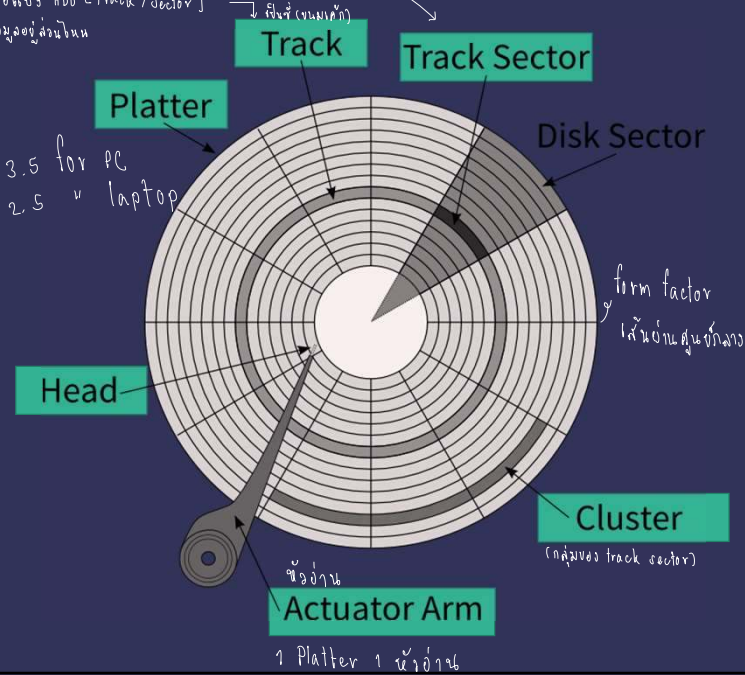
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Hard Disk

- The two most common form factors for modern HDDs are 3.5-inch, for desktop computers, and 2.5-inch, primarily for laptops
- Disk read/write heads are the small parts of a disk drive which move above the disk platter and transform the platter's magnetic field into electrical current (read the disk) or, vice versa, transform electrical current into magnetic field (write the disk).
- Formatting is the process of dividing the disk into tracks and sectors
- Revolutions per minute, RPM, is used to help determine the access time on computer hard drives. Most common RPM rates, in both laptop and desktop PCs, are between 5400 and 7200 RPM.

Revolution per minutes

format เพื่อแบ่ง HDD (Track / Sector)
เพื่อให้ง่ายต่อการจัดเก็บข้อมูล

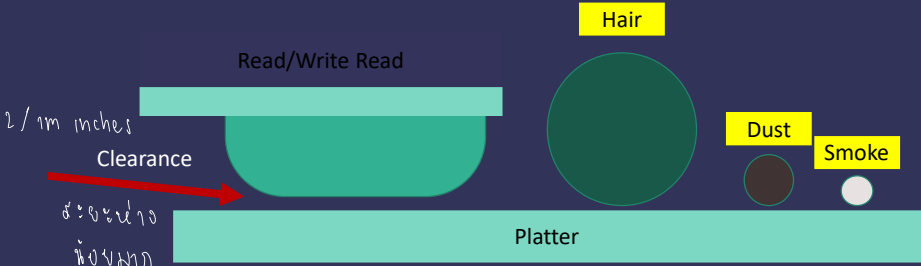


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Hard Disk

ข้อควรระวังเกี่ยวกับหัวอ่าน (อย่าได้เสียบของ)
↳ Defragment เพื่อป้องกัน

- A head crash occurs when a read/write head touches the surface of a platter
- Always keep a backup of your hard disk



ถ้าหัวเสียบ ก็เรียกว่า "Bad sector" ← มีวิธีการ clean disk เพื่อป้องกันหัวอ่านเป็น Bad sector เพื่อที่จะช่วยแก้ไขปัญหานี้

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Hard Disk

ถ้า ไขว้ → Mechanical Parts อย่าง Platter
เก็บไว้ → แคช cache
แล้ว → ถึงข้อมูลออกมา

Mechanical parts of hard disk directly affect the access time

Disk cache, sometimes called a buffer, consists of a memory chip(s) on a hard disk that stores frequently accessed data, instructions, and information

The larger the disk cache, the faster the hard disk

เก็บข้อมูลไว้ก่อน
Cache มา → ไว้

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Storage



Internal Hard Disk



Internal Solid-State Drive



External Hard Drive



Memory Cards



USB Flash Drive



Cloud Storage



Optical Discs



Network Attached Storage Device



Magnetic Stripe Card



Smart Card



RFID Tags



NFC Tags



Film

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Solid-state Drive

หน่วยความจำแบบแฟลช (non-volatile)
ตัวในเป็น IC (เรียกว่า หน่วย HDD)

- SSD is a flash memory storage device that contains its own processor to manage its storage
- An SSD has several advantages over traditional (magnetic) hard disks:

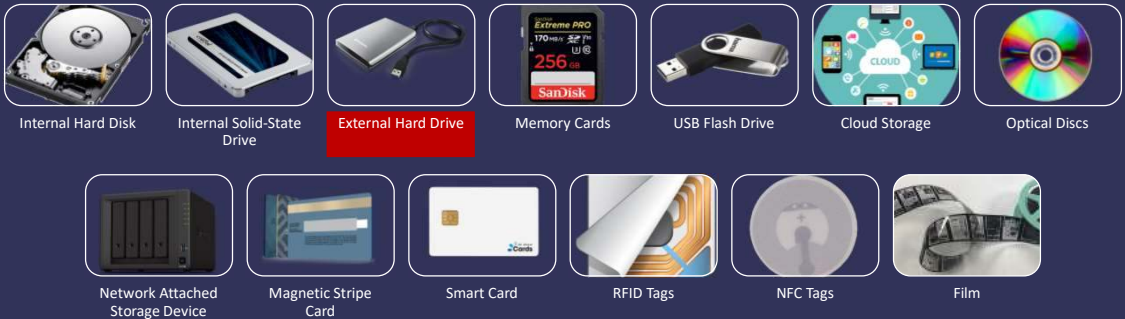
Pros

เร็วทันใจ Faster access times	เร็ว Faster transfer rates	เสียงเบา Quieter operation
ทนทาน More durable	น้ำหนักเบา Lighter weight	กินไฟน้อย Less power consumption
ไม่ร้อน Less heat generation	อายุการใช้งานนาน Longer life	ไม่ต้องจัดเรียงข้อมูล Defragmentation not required



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Storage



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External Hard Disk

HDD พกพา (เชื่อมต่อด้วย cable)

- An external hard disk is a separate freestanding storage device that connects with a cable to a USB port or other port on a computer or mobile device



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การจัดเก็บข้อมูลแบบจำลองเสมือน (รวม HDD / SSD หลายตัว มาเชื่อมต่อเป็นหนึ่งเดียว)

Redundant Array of Independent Disks

- RAID is a data storage virtualization technology
- It is a group of two or more integrated hard disks or SSDs that is for the purposes of data redundancy, performance improvement, or both.

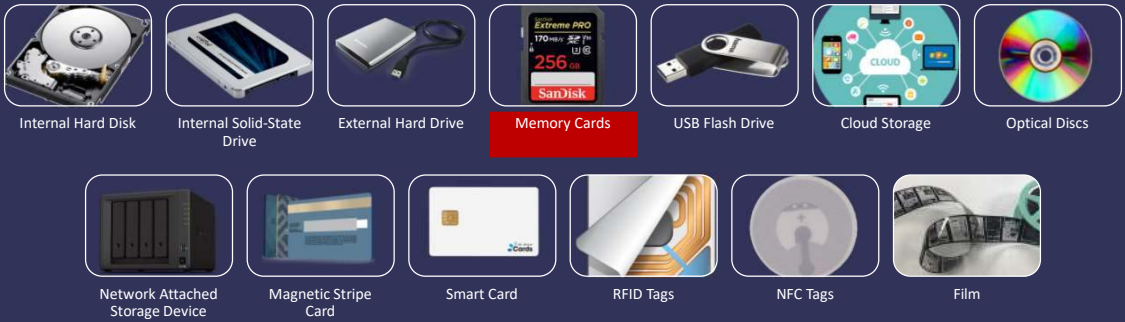
ปรับขนาดได้ด้วย software ที่ control by RAID



<https://en.wikipedia.org/wiki/RAID>

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Storage

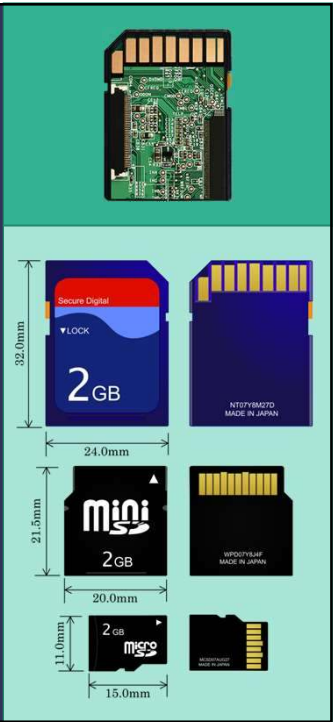


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Memory Card หน่วยด.จำแบบแฟลช

- It is a **removable flash memory** storage device that you insert and remove from a slot in a computer, mobile device, or card reader/writer

	Secured Digital (SD) Card	2 GB (Max)
	Secured Digital High Capacity (SDHC) Card	32 GB (Max)
	Secure Digital Extended Capacity (SDXC) card	2 TB (Max)
	Secure Digital Ultra Capacity (SDUC) card	128 TB (Max)



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Memory Card

Minimum sequential write speed	Speed Class		Corresponding video format				
	Speed Class	UHS Speed Class	Video Speed Class (new)	8K video	4K video	Full HD / HD video	Standard Video
Card image				The necessary speed varies by each recording/playback device condition, even in the same format.			
90MB/sec			V90	✓			
60MB/sec			V60	✓	✓		
30MB/sec		U	V30	✓	✓	✓	
10MB/sec	@	U	V10		✓	✓	✓
6MB/sec	@		V6		✓	✓	✓
4MB/sec	@					✓	✓
2MB/sec	@						✓

<https://www.kingston.com/en/solutions/personal-storage/memory-card-speed-classes>

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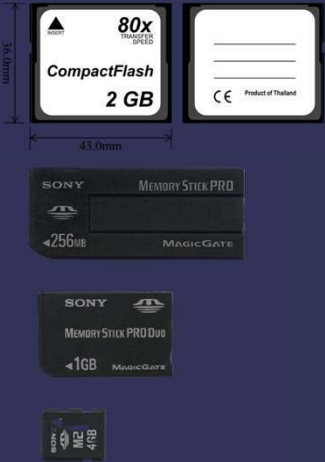
Memory Card

รองรับ Canon / Nikon

Compact Flash

Sony

Memory Stick



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Memory Card



Card Reader/Writer



Printer



Laptop



Smartphone/Tablet

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Storage



Internal Hard Disk



Internal Solid-State Drive



External Hard Drive



Memory Cards



USB Flash Drive



Cloud Storage



Optical Discs



Network Attached Storage Device



Magnetic Stripe Card



Smart Card



RFID Tags



NFC Tags

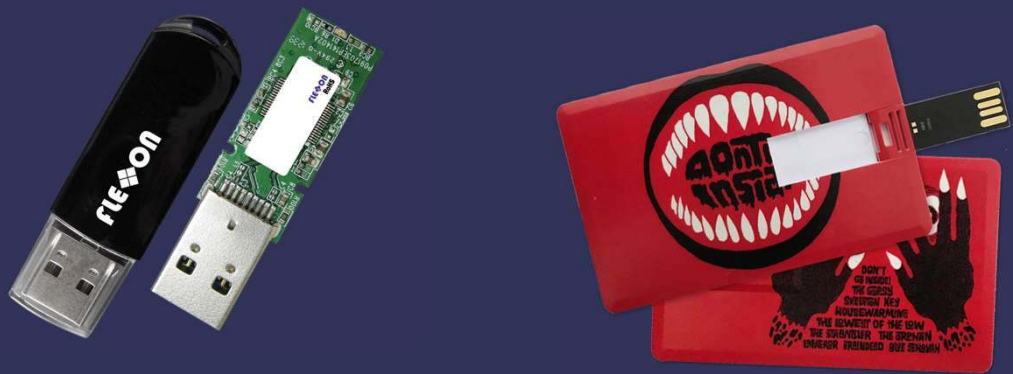


Film

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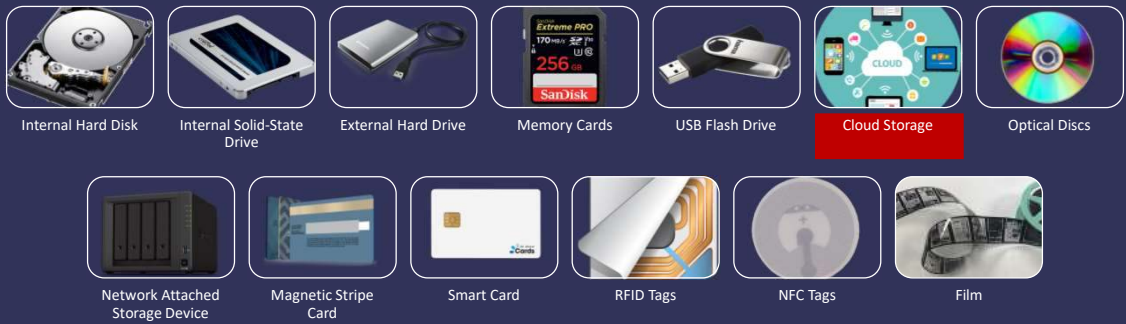
USB Flash Drive [thumb drive]

- USB flash drives plug into a USB port on a computer or mobile device



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Storage



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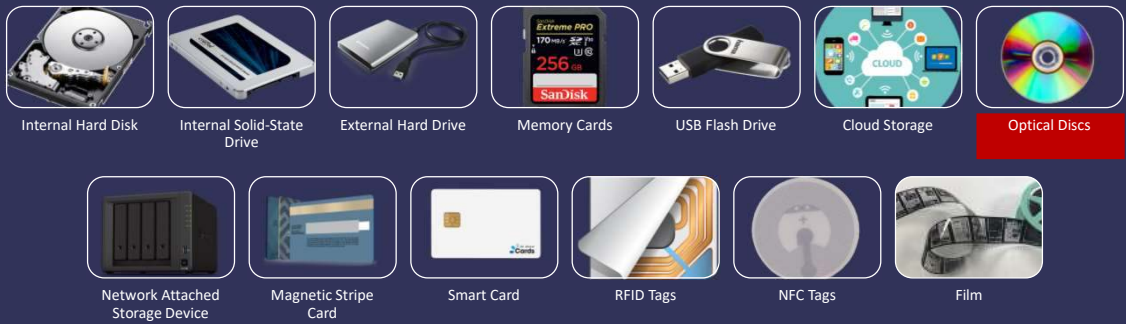
Cloud Storage

- Cloud storage is an Internet service that provides storage to computer or mobile device users



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Storage



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Optical Disc [Standard size 4.75 inches]

- An optical disc consists of a flat, round, portable disc made of metal, plastic, and lacquer that is written and read by a laser



COMPACT
disc
DIGITAL AUDIO

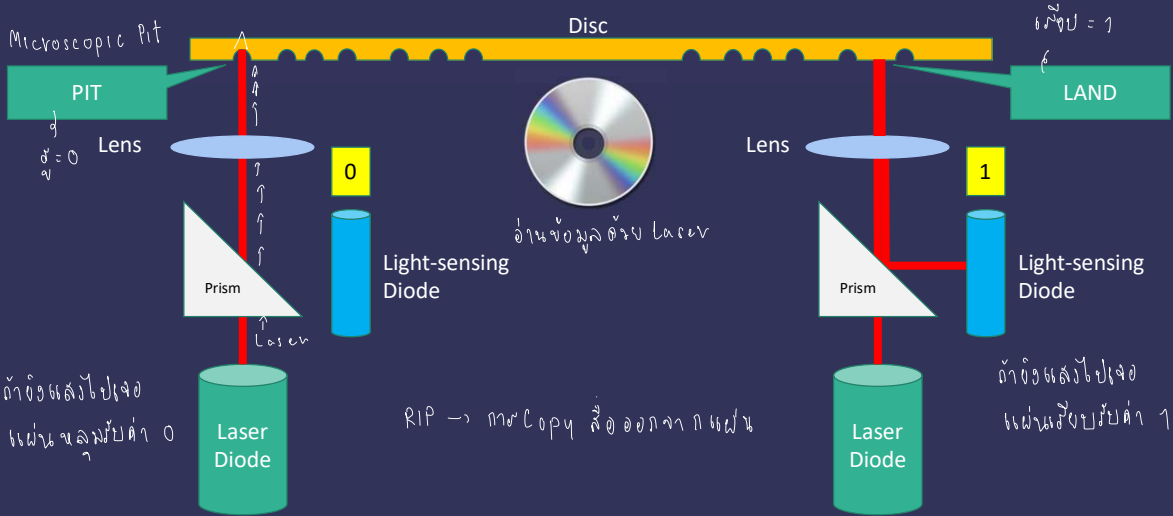
DVD
TM

HD DVD

Blu-ray Disc
TM

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How a Laser Reads Data on an Optical Disc



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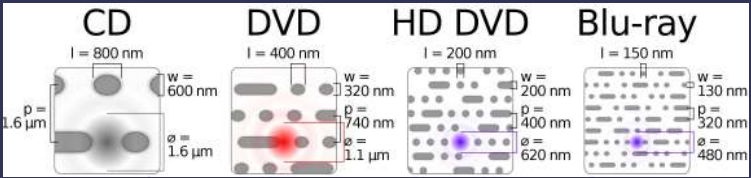
4.7 GB 700 MB
CD –vs– DVD



CD



DVD



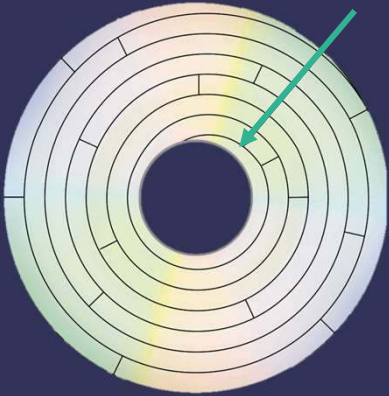
https://en.wikipedia.org/wiki/Optical_disc

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Optical Disc

- Optical discs commonly store items in a single track that spirals from the center of the disc to the edge of the disc
- Track is divided into evenly sized sectors

มีแค่ 1 แถบ วิ่งจากกึ่งกลางถึงขอบ
Single track spiral to edge of disc



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CD –vs– DVD

เขียนได้ 1 ครั้ง

CD

ลบได้ เขียนใหม่ได้

DVD

Read Only

- A CD-ROM can be read from but not written to
 - Single-session disc
- A CD-R is an optical disc on which users can write once, but not erase
- A CD-RW is an erasable multisession disc

- A DVD-ROM is a high-capacity optical disc on which users can read but not write on or erase
- A DVD-R or DVD+R are competing DVD-recordable WORM formats, on which users can write once but not erase
- DVD-RW, DVD+RW, and DVD+RAM are competing DVD-rewritable formats that users can write on multiple times

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Storage



Internal Hard Disk



Internal Solid-State Drive



External Hard Drive



Memory Cards



USB Flash Drive



Cloud Storage



Optical Discs



Network Attached Storage Device



Magnetic Stripe Card



Smart Card



RFID Tags



NFC Tags



Film

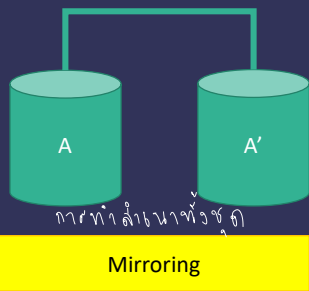
NAS (Network Attached Storage)

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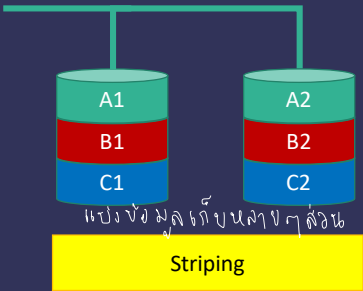
Enterprise Storage

- Enterprise hardware allows large organizations to manage and store data and information using devices intended for heavy use, maximum efficiency, and maximum availability
- It employs RAID to duplicate data, instructions, and information to improve **data reliability**

ทำ stock ข้อมูล



เก็บข้อมูลไว้สองที่

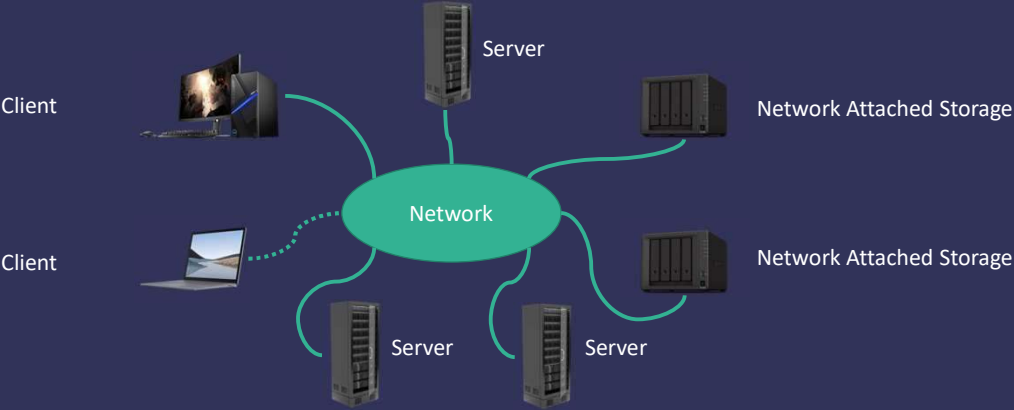


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เก็บข้อมูลของ User ในองค์กร (มี IP เป็นของตัวเอง)

Network attached storage (NAS)

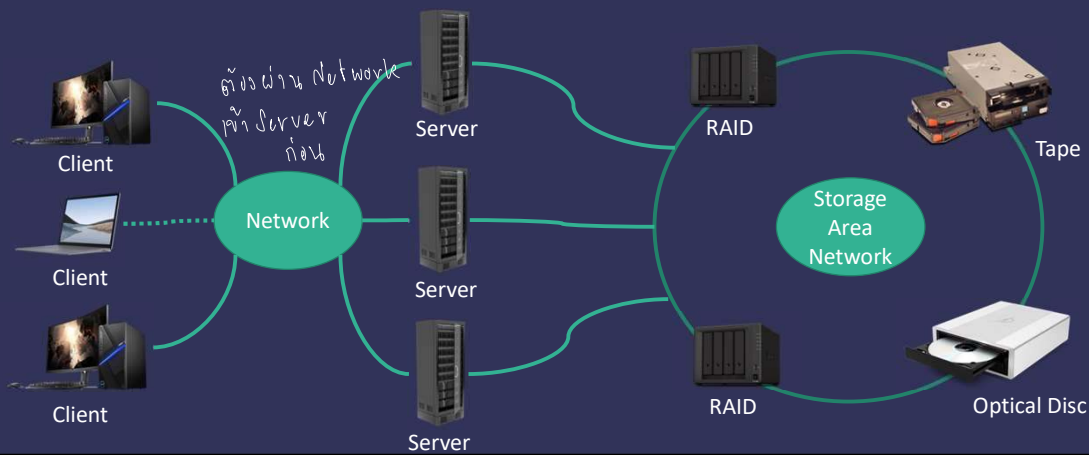
- NAS is a server that is placed on a network with the sole purpose of providing storage to users, computers, and devices attached to the network



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Storage Area Network (SAN)

- A storage area network (SAN) is a high-speed network with the sole purpose of providing storage to other attached servers



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Magnetic Tape System

ระบบสำรองข้อมูลที่ใช้เทปแม่เหล็ก

- Tape is a magnetically coated ribbon of plastic capable of storing large amounts of data and information
- A tape drive reads and writes data and information on a magnetic tape



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Storage



Internal Hard Disk



Internal Solid-State Drive



External Hard Drive



Memory Cards



USB Flash Drive



Cloud Storage



Optical Discs



Network Attached Storage Device



Magnetic Stripe Card



Smart Card



RFID Tags



NFC Tags



Film

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Other Types of Storage



Magnetic Stripe Card



Smart Card



RFID Tag

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ក. វិទ្យាសាស្ត្រស្តុកទិន្នន័យ

Near Field Communications

- An NFC-enabled device contains an NFC chip
- An NFC tag contains a chip and an antenna that contains information to be transmitted
- Most NFC tags are self-adhesive



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Film

ឯកសារដែលបានរៀបចំជាមុនត្រូវបានផ្ទេរទៅក្នុងស្រទាប់

- Microfilm and microfiche store microscopic images of documents on a roll or sheet film



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